08 · Roadmap and Analysis

Doc: LOT-HW-01-ROAD · **Rev:** A (2026-11-14) · Week 0 = the week Vadik approves decisions D-1 and D-2.

1. PHASES

Phase	Weeks	Deliverables	Exit gate
0 · Plan (done, session 1)	—	Docs 01–09, BOM, API spec, firmware skeleton, `lotctl` with mock server	Vadik signs off D-1…D-5
1 · Server API	0–2	`devices` + `device_notifications` migration, `/api/device/*` routes, Settings → Devices UI, `device_copy` in the Log tab, `device-nudge` job	`lotctl simulate --host https://lot-systems.com` passes against staging
2 · EVT-0 bench	0–3	Dev kits wired. Firmware drivers for LCD, haptic, BME688, camera, sleep. SoftAP Wi-Fi setup portal	"Coffee time!" arrives on the bench unit. Copy appears in Log
3 · EVT-1 design	2–6	KiCad schematic and layout, STEP enclosure, cover-glass drawings	DRC/ERC clean. PCBWay DFM review passes
4 · EVT-1 build	6–9	10 units (PCBWay quick-turn PCBA plus 5 CNC shells)	Power budget measured. RF ≥ −65 dBm closed. Qi charges through GL2
5 · DVT	9–15	Thin stack (if D-2 = 5 mm), 20 units, FCC pre-scan, drop and thermal tests	All of 01 §6 met on ≥ 18 of 20
6 · Certification	13–18	FCC Part 15B (+15C if chip-down), UN38.3 docs	Grant / test report in hand
7 · PVT / 100-unit run	16–22	110 units built, tested, packed. PDF manuals printed and in the box	≥ 95 % first-pass yield
8 · Field	22+	OTA channel, user feedback via Log, v2 planning (push socket, BLE app)	—

Critical path: **custom cell (BT1-DVT) and NCVM window (GL2)**, which have 4–6-week lead times. Order samples in week 0.

2. OPEN DECISIONS FOR VADIK

ID	Decision	Options	Recommendation
D-1	Wireless charging versus an all-steel back	NCVM glass window / all-steel with pogo dock / ceramic back	**NCVM glass window.** It keeps the silver look and Qi works (03 §1, 01 §3)
D-2	Thickness	6.5 mm with module (cheaper, faster, pre-certified) / 5.0 mm chip-down (+$10k FCC, +6 weeks)	Ship the first 100 at **6.5 mm**. Make 5.0 mm the v1.1 run
D-3	Camera purpose	Photo-to-Log only / also QR pairing / disabled until v2	Photo-to-Log on long press. QR pairing is a nice follow-on
D-4	Delivery model	Poll every 5 min (days of battery) / persistent socket (~1 day)	**Poll** with server-driven adaptive cadence
D-5	Steel grade	304 (cheaper, polishes well) / 316L (better corrosion, skin contact)	**316L** for a skin-contact pocket object

3. RISK REGISTER

#	Risk	L	I	Mitigation
R1	Steel enclosure kills Wi-Fi range	H	H	Antenna under the front glass, keep-out, tuning on EVT-1, chip antenna on DVT
R2	Qi heats the steel ring and charging stalls	M	H	Window ≥ coil + 3 mm, NTC at the cell, 1 C limit. Pogo-pin fallback footprint on the PCB
R3	5 mm unachievable with a camera	H	M	D-2: 6.5 mm first. Camera beside the battery in a PCB cut-out
R4	Memory LCD outline does not fit with the camera and button	M	M	Verify the LS011B7DH03 drawing in week 0. Alternative: 0.96" 128 × 64 bare OLED (higher power)
R5	Thin camera module sourcing (single supplier, EOL)	M	M	Qualify 2 module houses. Firmware supports GC032A and GC0308
R6	Custom cell MOQ above 100	H	M	Pay NRE, or use the catalogue 303030 for the first run
R7	FCC delay	M	H	Module path for the first run. Pre-scan at DVT
R8	Notification fatigue	M	M	Server caps: 8/day, 45-min gap, quiet hours, circadian phase
R9	Token leak from a lost device	L	M	Revoke in Settings. Device scope is write-only `device_*` logs plus own notifications

4. ANALYSIS: IS THE CONCEPT SOUND?

- **Strong fit with LOT.** The site already produces proactive, context-aware prompts (Memory Engine questions, self-care, circadian phase). TILE is a *physical, quiet output channel* for them, and the Copy button closes the loop into the Log, which is LOT's record of the day.
- **The hard parts are mechanical and RF, not software.** Steel, a 5 mm thickness and wireless charging pull against each other, and §2 resolves them.
- **The software is low-risk:** about 8 new routes, 2 tables and 1 scheduled job, plus the Log allow-list change. The mock server in `lotctl` already pins the contract.
- **Unit economics at 100:** about $153 build cost plus about 40*perunitamortisedcertification*, *soabout* * *190–$200 landed per unit**. Fine for a limited, numbered "LOT TILE 001/100" edition. At 1 000 units this drops below about $70.

5. BUDGET SUMMARY (100-UNIT EDITION, USD ESTIMATES)

Item	Estimate
Dev kits and EVT-0	300
EVT-1 (10 units, quick-turn)	2 500
DVT (20 units)	4 000
Certification (module path)	4 000
PVT run (07 §3)	15 300
Contingency 15 %	3 915
Total programme	**≈ 30 000**

6. NEXT SESSION (SESSION 2) PLAN

1. Implement phase 1 on the server: migration, `src/server/routes/device-api.ts`, the auth hook, `device_copy` in `LogEvent` and `displayableEvents`, then run `lotctl simulate` against a local server.
2. Start the KiCad schematic in `hardware/lot-tile/pcb/`.
3. Firmware: Memory LCD driver and DRV2605L pager waveform (needs a toolchain-enabled environment to compile).
